Ganesan Sai Krishnan

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.85
Narayana Junior College	Class XII (Telangana State Board of Intermediate Education)	2017	98.30%
Balaji High School	Class X (Board of Secondary Education Telangana State)	2015	9.50%

ACHIEVEMENTS & RECOGNITIONS

- Received the **Certificate of Academic Excellence** from Vardhaman College of Engineering and **Cambridge English Level-1 (Business Vantage)** certification.
- Received the **ACE Award - FY25 Extra Mile Award** and **ATCI NA Q1-FY25 Pinnacle Award - NorthStar (Individual)** for impactful contributions, operational excellence, continuous learning, and leadership.
- Received the **Stellar Award - Sep'24** for contributions to **cloud automation, dashboard migrations, and AWS cost optimization**.

MASTER'S THESIS

Title: Cellulose Paper-Supported CuI/Fe2O3 p-n Heterojunction Photodetector

- **Materials & Device Innovation:** Designed and fabricated a CuI/Fe2O3 p-n heterojunction photodetector on an eco-friendly cellulose paper substrate, using cellulose as a lightweight, cost-effective, and flexible alternative to conventional rigid glass/silicon
- **Process Optimization & Interfacial Control:** Systematic SILAR process control optimized thin-film adhesion and interfacial properties between heterojunction layers on paper, successfully resolving substrate-film integration challenges
- **Broadband Photodetection Performance:** Demonstrated a self-assembled heterojunction architecture capable of high-performance photo-response across the UV-Visible-NIR spectrum with bias-tunable gain.
- **Publication & Scientific Communication:** Authored a peer-reviewed manuscript documenting fabrication protocols, electrical characterization data, and photodetection mechanisms.

PROJECTS

16×16 6T SRAM Array Design

- Designed and analyzed a **16×16 SRAM array** consisting of **256** conventional **6T SRAM cells** with organized word lines and differential bit-lines
- Studied the **6T SRAM cell architecture**, including cross-coupled CMOS inverters, access transistors, and transistor sizing for reliable read/write operation
- Designed and analyzed **peripheral circuits** including a **4-to-16 row decoder, sense amplifiers, write drivers, and bit-line precharge/equalization circuits**
- Evaluated **performance, power, stability, and reliability**, focusing on access time, power consumption, Static Noise Margin (SNM), process/voltage/temperature variations, and layout area

Design and Implementation of 32-bit ALU using Memristor Modelling

- Designed and implemented **a 32-bit ALU** using **memristor-based logic** for low-complexity and nanoscale digital circuit implementation
- Developed and modelled **memristor-based logic gates** including AND, OR, NAND, XOR, along with arithmetic blocks such as the **half adder**
- **Simulated and validated** memristor-based logic circuits through circuit-level waveform analysis to verify functional operation
- **Compared transistor count and power characteristics** of conventional transmission-gate, CMOS, and GDI-based

32-bit ALU architectures to evaluate the benefits of memristor-based designs

High-Speed and Low-Power 2's Complement Circuit

- Designed a high-speed and low-power 2's complement circuit using a seek-method architecture to simplify binary complementation
- Developed scalable 2-bit, 3-bit, 4-bit, and n-bit architectures, using XOR-based logic to perform 2's complement generation without the conventional inverter-and-adder structure
- Implemented and functionally verified the proposed circuit using Cadence Virtuoso with 45 nm technology libraries across all possible input combinations
- Analysed and compared power dissipation and propagation delay for 4-bit, 8-bit, and 16-bit implementations against existing 2's complement architectures
- Demonstrated reduced power and delay characteristics compared with multiplexer-based implementations, with reported reductions across 4-bit, 8-bit, and 16-bit designs

SKILLS

Language: HDL, Verilog, C, Python, Linux

Skills: RTL Design, FSM Design, Digital Logic Design, CMOS Design

Verification: Testbench Development, Functional Verification, Simulation & Debugging

Tools: Xilinx Vivado, Cadence Virtuoso, LtSpice, EDA Playground

RELEVANT COURSES

Digital Domain: Digital IC design, Introduction to HDL, Memory Device Technology for AI/ML, Digital IC design Lab, CMOS Digital Design, **Analog domain:** Analog IC design, Analog IC design Lab, **Technology & devices:** VLSI Technology, Nanoelectronics: Principles & Devices, Mesoscopic Device Electronics, Semiconductor device modeling.

POSITIONS OF RESPONSIBILITY

- **Teaching assistant** for **VLSI Technology** course by **Dr. Sushmee Badhulika**
- **Class Representative - M.Tech:** Represented the class and coordinated communication between students and faculty
- **Head Boy - School:** Served as Head Boy for two consecutive academic years (2013-14 and 2014-15)

WORK EXPERIENCE

- **Software Engineer at Accenture** with **3.5+ years of experience (Oct 2021 - Jul 2025)** in **cloud automation, data engineering, and operational solutions**
- **Led a team of 3 members** and mentored new joiners across multiple applications, managing task allocation, deliverables, onboarding, and knowledge transition
- Worked with **AWS, Databricks, Apache NiFi, Kibana, Grafana, MySQL Workbench, PyCharm, and DBeaver**, while also coordinating organizational events such as Town Halls and Leadership Connects

EXTRACURRICULAR

Interests: Singing Carnatic Music & Bhajans, Badminton, Swimming